HackInsight 2026: Lahore Air Quality Decision System

Executive One-Page Summary | Track: Air Quality | Team H4krz (Data-to-Decision)

Team Members: M. Saad Ahmed, Arbish, Hasan Shahir, Noman Khursheed | Verified Test Suite: 52/52 Passed

1. Problem Definition & Primary User Decision (Guide Ch. 3 & 5)

- Problem: Severe winter PM2.5 smog in Lahore (Oct-Feb) causes overnight spikes exceeding 250 ug/m3, risking child health.
- Primary User & Decision: A school headmaster / administrator in Lahore deciding at 4:00 PM PKT the day prior whether to cancel, modify, or move morning outdoor assemblies and physical activities based on reliable next-day PM2.5 forecasts.
- Statistical Rules: Midnight-to-midnight PKT day definition (UTC+5), strict 18-of-24 valid hours completeness rule, and deterministic 16-character SHA-256 primary keys (record_id) ensuring 100% reproducibility across all warehouse layers.

2. Warehouse & Data Quality Agent (Ch. 4, 6)

- Data Volume: 86,363 regular hourly rows, 65,867 valid PM2.5 observations (>50,000 threshold passed) across 8 verified Lahore stations.
- Independent Sources: OpenAQ v3 (AirGradient & Hawanama) and Open-Meteo ERA5 Reanalysis & Forecast. 100% CC BY 4.0 compliant.
- Licensing Contract: Station ieee_punjab_univ strictly excluded from cleaned/processed marts due to unverified licensing.
- DuckDB Warehouse: 3 layers (raw JSONs, cleaned hourly Parquet, daily.parquet mart) synced to data/warehouse.duckdb via safe DataFrame registration.
- SHA-256 Ledger: manifest.csv audits 481 physical files (466 raw evidence JSONs).
- Data Quality Agent: Validates Pandera bounds and nulls; executes fix (unit correction), fill (linear interpolation), or flag (anomalies);
- Staleness Resilience: Logs 666 quality actions to quality_log.csv. Late sensors trigger non-fatal WARN_STALE, keeping pipeline alive.

3. Predictive Model & Group Fairness (Ch. 7, 8)

- Declared Baseline: Persistence (tomorrow equals today), MAE: 13.593 ug/m3.
- Final Model: HistGradientBoosting Regressor (residual delta framing). Test MAE: 12.594 ug/m3 (+7.35% gain; declared >=15% target NOT met; docs state honestly).
- Temporal Split: 351 train dates (1,587 rows), 75 val dates (367 rows), 76 test dates (453 rows). Zero future leakage; untouched test set.
- 80% Uncertainty Bounds: Multi-quantile pinball loss models (10th & 90th percentiles) yielding 76.8% empirical test coverage.
- Group Fairness Audit (reports/model_errors_by_group.csv):
  * Stations: Sandha Road (10.98 MAE, +1.7%), Model Town (12.61 MAE, +8.7%), Trinity School (11.47 MAE, +22.8%).
  * Seasons: Non-Smog 12.13 MAE (+6.0%) vs Smog Season 12.13 MAE (+6.0% gain where smog warning matters most).
- 3-Step Optimization Log: Iter 1 (HistGradientBoosting_DirectTarget, -1.3%); Iter 2 (HistGradientBoosting_ResidualDelta, +7.3% [selected]); Iter 3 (HistGradientBoosting_CityTrendL2, +6.1%).

4. Insight Agent (RAG) & Pakistan Bonus (Ch. 9)

- Hard Rule 1 (No Invented Numbers): Every numerical value is pulled directly from gold tables; references record_id row IDs.
- Hard Rule 2 (Cited Sources): Grounds recommendations in WHO 2021 Global Air Quality Guidelines and EPA Punjab Smog Action Plan.
- Hard Rule 3 (No Synthetic Evaluation): Evaluated on real held-out observations; tests pass with 0 hallucinations.
- Honest Abstention: Returns 'I don't know based on the available evidence' on out-of-scope queries (e.g. crop yield or traffic).
- Pakistan Impact Bonus (+10 pts):
  * Urdu Output Toggle: Bilingual support and translations (Urdu Tarjuma) for school staff.
  * Local Validation: EPA Punjab Smog Action Plan thresholds (<50 normal, 50-150 sensitive exertion, 150-250 cancel outdoor sports, >250 closure).
  * Emergency Health Disclaimer: Clear referral to official hospitals.

5. Hourly Pipeline & Business Model (Ch. 10, 11)

- Hourly Automation (Prefect): Robust scheduler fires at :05 past every hour. Consecutive SUCCESS runs logged in logs/prefect_runs.csv.
- Live Dashboard (Streamlit): 3 tabs (Live Monitoring, Model & Fairness, RAG Assistant) displaying freshness badges (FRESH / LATE), DQ audit, and last run.
- Primary User vs Customer:
  * Users: School administrators, sports directors, and clinic staff.
  * Payers: School district boards, private school chains, and municipal bodies.
- Revenue Model: Freemium SaaS. Free public portal for citizens; $25/mo per school subscription for automated SMS/email alerts and risk auditing.
- Operational Cost: ~$15/mo (serverless compute & free-tier OpenAQ/Open-Meteo APIs). Extremely low overhead with >20x value return.
- Value Generated: Prevents severe asthma attacks in students, avoids last-minute bus/event cancellations, and saves 3.5 hrs/week per school.

GitHub: https://github.com/SaadAhmed8/h4krz-hackinsight-2026 | HF Space: https://hf.co/spaces/hasanshahir/lahore-air-quality-decision-system
Reproducibility: 'make data' or 'python -m src.pipeline' rebuilds all layers | Hourly Automation: 'python -m src.prefect_pipeline'
Integrity Guarantee: Zero hard-coded values, 100% CC BY 4.0 licences, 52/52 automated unit & integration tests passing.